



SIMATS Online Programming Lab

JAVA PROGRAMMING



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CO2-AT1-Assignment1

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QUESTIONS

Q1

Smart Document Editor using String, StringBuilder and StringBuffer

10 marks

**Q2**

Hospital Token Management System using List Interface

10 marks

**Q3**

Airport Boarding Queue Management using Queue Interface

10 marks

**Q4**

Digital Certificate Verification using Set Interface

10 marks

**Q5**

Smart Inventory Management using Map Interface

10 marks



5/5 answered

Q2

Hospital Token Management System using List Interface

10 marks

Hospital Token Management System using List Interface

Store patient details using an appropriate List implementation.

Register new patients.

Display patients in registration order.

Update patient information using Patient ID.

Remove cancelled appointments.

Display all patients of a particular department.

Find the first and last registered patients.

Requirement: Use a suitable List implementation

Your Code

```
1 import java.util.ArrayList;
2 import java.util.List;
3 import java.util.Scanner;
4
5 public class HospitalTokenManagement {
6     static class Patient {
7         int id;
8         String name;
```

```
9      String department;
10
11      public Patient(int id, String name,
12                     this.id = id;
13                     this.name = name;
14                     this.department = department;
15      }
16      @Override
17      public String toString() {
18          return "ID: " + id + ", Name: "
19      }
20  }
21  public static void main(String[] args) {
22      Scanner scanner = new Scanner(System
23      List<Patient> patients = new ArrayLi
24
25      if (!scanner.hasNextInt()) {
26          return;
27      }
28      int count = scanner.nextInt();
29      for (int i = 0; i < count; i++) {
30          int id = scanner.nextInt();
31          String name = scanner.next();
32          String dept = scanner.next();
33          patients.add(new Patient(id, nam
34      }
35      System.out.println("All Registered P
36      for (Patient p : patients) {
37          System.out.println(p);
38      }
39      if (scanner.hasNextInt()) {
40          int updateId = scanner.nextInt()
41          String newName = scanner.next();
42          String newDept = scanner.next();
43          for (Patient p : patients) {
44              if (p.id == updateId) {
45                  p.name = newName;
46                  p.department = newDept;
47                  break;
48              }
49          }
50      }
51      if (scanner.hasNextInt()) {
52          int cancelId = scanner.nextInt()
53          patients.removeIf(p -> p.id == c
```

```
54     }
55     if (scanner.hasNext()) {
56         String searchDept = scanner.next
57         System.out.println("\nPatients i
58         for (Patient p : patients) {
59             if (p.department.equalsIgnor
60                 System.out.println(p);
61         }
62     }
63 }
64 if (!patients.isEmpty()) {
65     System.out.println("\nFirst Regi
66     System.out.println("Last Registe
67 }
68
69 scanner.close();
70 }
71 }
```

Input

```
3
101 Alice Cardiology
102 Bob Neurology
103 Charlie Pediatrics
```

Output

```
All Registered Patients:
ID: 101, Name: Alice, Department: Cardiology
ID: 102, Name: Bob, Department: Neurology
ID: 103, Name: Charlie, Department:
Pediatrics
```



Run



Save